

BME 5313 “Cellular Mechanobiology”

Dr. Eric G. Meyer

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Class hours: 5-6:15 PM Tues and Thurs

Office hours: T 11am or R 2pm
or by appointment

Office: J337

Place: E213

Prerequisite(s): Graduate level. Knowledge of undergraduate level mechanics of solid mechanics of materials, biology and physiology.

Course Content

BME 5313 covers the biochemical signaling in response to various mechanical stresses in the context of physiology and pathophysiology. Topics include the behavior of live cells during cell motility, force generation, and interaction with the extracellular matrix; the advanced biomechanical testing tools used for in vitro characterization of living cells; mechanotransduction that converts mechanical forces into biochemical signaling.

Course instructional format

The class will have lectures and class discussion involving student participation. Students will be assessed based on occasional homework assignments, laboratory projects, in class presentations, a term paper and two exams. This is a three credit class consisting of two meetings per week and directed learning.

Required Textbook

Introduction to Cell Mechanics and Mechanobiology

By; Jacobs, Huang, and Kwon ISBN 978-0-8153-4425-4

Grading

- **20% Homework**
 - Students may discuss homework problems, but should write up solutions independently. Late submissions will not be accepted, unless previously arranged with the instructor.
- **10% Projects**
 - Groups will complete experimental projects in the laboratories. Reports will include the background motivation, methods, results and conclusions of the project. Additional details will be provided during the semester.
- **15% Journal Club**
 - In class presentations on scientific journal articles to summarize the hypothesis, methods and lead a discussion of the results and conclusions. All students are required to read the article being discussed for each journal club presentation. Scheduled topics will be provided during the semester.
- **15% Term Paper**
 - Individual research on a topic of the students' choosing. Additional details will be provided later in the semester.
- **40% Exams**
 - A midterm and a final exam will assess the material covered in the first and second parts of the course, respectively.

Grading Scale:

Grades will be assigned as follows:

– A	≥ 90%
– A-	88.5-90%
– B+	86.9-88.4%
– B	80-86.8%
– B-	78.5-79.9%
– C+	76.9-78.4%
– C	70-76.8%
– C-	68.5-69.9%
– D+	67.1%-68.4%
– D	60-66.9%
– F	<60%

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Violations of academic ethics will be handled according to University policies as set forth in the University's Academic Honor Code, included here by reference (see http://www.ltu.edu/currentstudents/honor_code.asp). Specific policies for assignments in this course are:

- **Quizzes and papers:** All quizzes and exams must be by individual effort. Unprogrammed calculators are allowed, but use of laptops or programmed calculators as well as observation of others' work will be considered "cheating" as defined in the Honors Code. Plagiarism will not be tolerated.

Note: Unforeseen circumstances may necessitate changes, additions, or deletions to this syllabus during the term.

Semester schedule – Instructor reserves the right to modifications as needed

Class	Date	Topic	Reference	Assignment
1	8/25	LN1 Concepts	Ch1	Lit Review
2	8/27	LN2 Origins	Thompson	JC
3	9/1	LN3 Innovations		
4	9/3	LN4 Cell Biology Review	Ch2	HW1
5	9/8	LN5 Growth	Gilbert	
6	9/10	LN6 Growth Cont.		
7	9/15	LN7 Solid Mechanics Review	Ch 3	HW2
8	9/17	LN8 Solid Mech. Cont.		
9	9/22	LN9 Fluid Mechanics Review	Ch4	HW3
10	9/24	LN10 Fluid Mech. Cont.		
11	9/29	Guest seminar		
12	10/1	LN11 Statistical Mechanics	Palsson	HW4
13	10/6	LN12 Statistical Mech. Cont.	Ch5	
14	10/8	Cont...		
15	10/13	Exam 1		
16	10/15	Laboratory Methods 1		Midterm
17	10/20	LN13 Cellular Polymers	Ch6	Project 1
18	10/22	LN14 Cellular Poly. Cont.	Ch7	
19	10/27	LN15 Cellular Networks		HW5
20	10/29	LN16 Cellular Net. Cont.	Ch8	
21	11/3	LN17 Cell Membrane		
22	11/5	LN18 Cell Memb. Cont.	Ch9	HW6
23	11/10	LN19 Cell Migration		
24	11/12	LN20 Cell Contraction	Ch10	Term Paper
25	11/17	LN21 Functional Tissue Engr.		
26	11/19	Laboratory Methods 2	Mow	
27	11/24	LN22 Mechanotransduction		Project 2
28	12/1	LN23 Biochemical Regulation	Ch11	
29	12/3	LN24 Mechanoregulation	Carter	
Final	TBD	Exam 2		Final

Additional Reference Materials

What Is Intelligence? By Blaise Agüera y Arcas, 2024. DOI 10.1162/ANTI.5CZB. Free online

<https://whatisintelligence.antikythera.org/>

On Growth and Form By; D'Arcy Wentworth Thompson. Out of copyright, many versions available. Free download <https://archive.org/details/ongrowthform00thom>

Developmental Biology By Scott Gilbert and Michael Barresi, 2019. ISBN 978-1605358222

Systems Biology: Constraint-based Reconstruction and Analysis By; Bernhard Palsson, 2015. ISBN: 9781107038851

Basic Orthopaedic Biomechanics & Mechano-biology By; Van C. Mow, Rik Huiskes, 2005. ISBN 0-7817-3933-0

Skeletal Function and Form By; Dennis Carter and Gary Beaupre, 2007. ISBN: 9780521714754

Invention and Evolution: Design in Nature and Engineering By; M. J. French, 1988. ISBN: 0521307599

Various journal articles and handouts will be assigned as needed